

# llmXive Automated Review of ABot-N1: Toward a General Visual Language Navigation Foundation Model

*llmXive automated review panel*

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**Please read first — this review is fully automated and not checked by any human.** It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

## How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

**This paper's panel (9 lenses):** Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

## Claim Accuracy

*Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.*

**Reviewer model:** [qwen.qwen3.5-122b](#)

**Prompt:** `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a robust set of claims supported by internal tables, but there are minor inaccuracies in how the abstract and text summarize the numerical results, specifically regarding the distinction between percentage points and relative percentages, and the conflation of different metric definitions.

In the Abstract, the claim “boosting POI arrival by 35.0% (to 77.3%)” is mathematically ambiguous. The data in Table 4 shows an increase from 42.3% to 77.3%, which is an absolute increase of 35.0 *percentage points*. Stating “by 35.0%” typically implies a relative increase (which would be ~82%), potentially misleading the reader. This should be corrected to “35.0 percentage points” to align with standard scientific reporting.

Additionally, the Abstract states “95.4%/92.9% SR in complex indoor and outdoor scenes” as if these are comparable metrics. However, Table 1 (Indoor) uses a strict zero-collision criterion ( $SR < 1col$ ), while Table 2 (Outdoor) uses a relaxed criterion allowing up to three collisions ( $SR < 3col$ ). These are not directly comparable success rates. The abstract should explicitly qualify these numbers to reflect the different evaluation protocols (e.g., “95.4% SR indoors (strict) and 92.9% SR outdoors (relaxed)”).

Finally, in Section 6.1.5, the text claims ABot-N1 surpasses “all baselines” with the “sole exception of STT TR”. While true that it loses STT TR to Qwen-RobotNav-4B, the phrasing “surpassing... all baselines” could be misinterpreted as a blanket statement. It is more precise to state it achieves the best SR across all splits and the best TR on DT/AT, while noting the specific 0.2% gap on STT TR against the Qwen baseline. These are minor editorial fixes to ensure the claims precisely match the evidence provided in the tables.

**Action items.**

- **[writing]** Abstract claims ‘95.4%/92.9% SR’ conflating distinct metrics. Table 1 (indoor) uses strict  $SR < 1col$ , while Table 2 (outdoor) uses relaxed  $SR < 3col$ . Rephrase to specify ‘95.4% SR (indoor, strict) and 92.9% SR (outdoor, relaxed)’ to avoid implying comparability.
- **[writing]** Abstract states ‘boosting POI arrival by 35.0% (to 77.3%)’. Table 4 shows an increase from 42.3% to 77.3%, which is +35.0 percentage points, not a relative 35% increase. Clarify to ‘by 35.0 percentage points’ to prevent misinterpretation.
- **[writing]** Section 6.1.5 claims ABot-N1 ‘surpasses all baselines’ except STT TR. Table 5 shows Qwen-RobotNav-4B (90.0%) beats ABot-N1 (89.8%) on STT TR. The phrasing is slightly ambiguous; clarify that it leads on SR across all splits but trails Qwen-RobotNav-4B on STT TR by 0.2%.

**Figure Critic**

*Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_figure_critic.md`

### Figure 1

The figure provides a clear overview of the dataset statistics and pipeline stages, but the visual representation of Stage 3 omits the occupancy grid and A\* algorithm mentioned in the caption, relying instead on a ‘Normal Prior’ label that is not fully integrated into the flow.

### Figure 2

Figure 2 effectively illustrates the four key segments of the point-goal deployment with clear visual examples. However, the connection between the ‘Slow System Reasoning’ outputs and the agent’s actual execution in the VLA/Third-View panels is not explicitly visualized, and the composite nature of the figure could be better structured to emphasize the continuous episode.

### Figure 3

Figure 3 effectively illustrates three object-goal deployment cases (outdoor bench, indoor chair with bottle, fire extinguisher) with clear visual overlays for CoT reasoning, affordance pixels, and target pixels. The layout is organized, the text is legible, and the caption accurately describes the content shown.

### Figure 4

The figure effectively visualizes the robot’s navigation in three distinct POI scenarios with clear tri-view overlays, but the caption’s description of the third case (Luckin Coffee) contradicts the visual content by mentioning obstacles not present in the image.

### Figure 5

Figure 5 effectively illustrates the slow-system reasoning process with clear visualizations of affordance pixels and CoT. However, the target pixel visualization is inconsistently applied across the four critical moments, appearing only in the final panel despite the caption’s claim, and the legend is missing from the first three panels.

### Figure 6

Figure 6 effectively visualizes the person-following deployment across outdoor and indoor scenarios. The figure clearly distinguishes between raw observations and model outputs (Target Pixel, Affordance Pixel) using consistent color coding and labels, and the caption accurately describes the depicted tasks.

## Figure 7

The figure provides a clear visual overview of the ABot-N1 architecture and performance, but the caption mentions ‘EVT-Bench’ which is not explicitly labeled in the radar chart, and the radar chart legend is too small to read clearly.

## Figure 8

The figure effectively visualizes the dual-system architecture with clear data flow, but the caption is truncated at the end. Additionally, several icons (hourglass, lightning, fire emojis) are used without definition in the caption or a legend.

## Figure 9

The figure effectively visualizes the data pipeline and composition, but the main donut chart suffers from poor legibility due to rotated text and a confusing inner ring where the ‘Object Goal’ segment is visually merged with its neighbor. Additionally, the total count in the center does not match the sum of the labeled segments.

## Figure 10

The figure effectively visualizes the data pipeline and a sample output, but the caption lacks specific definitions for the coordinate values shown in the sample and could be more precise about the target pixel annotations.

## Figure 11

The figure effectively visualizes the instruction decomposition and alignment process, but the example sample on the right fails to show the target pixel annotations mentioned in the caption, and the verification stage described in the text is not explicitly illustrated in the diagram.

## Figure 12

The figure effectively visualizes the data pipeline and sample structure, but the caption’s specific claim about A\* filtering is not explicitly labeled in the diagram, and the structured sample text is cluttered with raw coordinate syntax.

**Action items.**

- **[science]** Figure 1: The caption states that Stage 3 uses ‘A\* on MoGe-V2-derived 2D occupancy grids’, but the diagram explicitly labels the grid generation step as ‘MoGe-V2 Normal Prior’ and does not visually depict the occupancy grid or the A\* algorithm, creating a disconnect between the text description and the visual pipeline.
- **[writing]** Figure 1: The ‘Stage 3’ section contains a floating label ‘MoGe-V2 Normal Prior’ pointing to a pyramid icon without a connecting box or clear definition of its role in the ‘Traversability-Aware Sampling’ flow, making the diagram’s logic harder to follow.

- **[writing]** Figure 2: The caption describes the content as ‘Four segments of a long-range outdoor episode,’ but the figure itself is a composite of four distinct sub-figures (Seg1-Seg4) without a unifying visual frame or explicit labels connecting the segments into a single continuous narrative.
- **[science]** Figure 2: The ‘Slow System Reasoning’ panels display ‘Affordance Pixel’ markers (green dots) on the ground, but the corresponding ‘VLA’ and ‘Third-View’ panels below them do not show these markers, making it difficult to verify if the visual guidance aligns with the agent’s actual perception or trajectory.
- **[science]** Figure 4: The caption claims ‘slope navigation and staircase avoidance’ for the Luckin Coffee case, but the visual evidence shows the robot navigating a flat sidewalk and avoiding a red barrier; no slope or staircase is visible.
- **[writing]** Figure 4: The ‘Slow System Reasoning’ row contains red, blue, and green bounding boxes, but no legend or caption text explains what these specific colors signify (e.g., confidence levels, specific reasoning steps, or error types).
- **[science]** Figure 5: The ‘Target Pixel’ (red star) is shown in the final ‘bar approach’ panel but is absent from the first three ‘Slow System Reasoning’ panels despite the caption claiming ‘target pixel visualizations’ are present at all four critical moments.
- **[writing]** Figure 5: The legend in the ‘bar approach’ panel labels the red star as ‘Target Pixel’ and the green dot as ‘Affordance Pixel’, but the ‘Slow System Reasoning’ panels for stair descent, gym entry, and gym exit only show the green dot without a legend, creating ambiguity about the red star’s absence.
- **[science]** Figure 7: The caption claims evaluation on ‘EVT-Bench’, but the radar chart legend and axis labels only show ‘ABoT-N1’, ‘Previous BOTA’, ‘NavFoM’, ‘Uni-NavVid’, and ‘Qwen-VLA’, with no explicit reference to EVT-Bench or its specific metrics.
- **[writing]** Figure 7: The legend for the radar chart is extremely small and illegible, making it difficult to distinguish between the different model lines (e.g., ‘Previous BOTA’ vs ‘NavFoM’).
- **[writing]** Figure 8: The caption text is truncated at the end (‘predict continuo’), cutting off the description of the MLP output.
- **[science]** Figure 8: The ‘Slow System’ block contains a purple hourglass icon and fire emojis, but the caption does not define these symbols or explain their significance (e.g., latency, compute intensity).
- **[science]** Figure 8: The ‘Fast System’ block contains a lightning bolt icon and a fire emoji, which are undefined in the caption and lack a legend.
- **[science]** Figure 9: The inner ring of the main donut chart contains a segment labeled ‘Object Goal 0.1M’ that is visually indistinguishable from the adjacent ‘POI Goal 3.0M’ segment due to identical coloring and lack of a visible boundary line, making the chart unreadable.
- **[science]** Figure 9: The ‘TOTAL 30M’ label in the center of the main donut chart is mathematically inconsistent with the sum of the visible segment labels.

$(6.2+2.4+2.7+3.0+0.1+0.1+5.1+6.8+3.4 = 29.8\text{M})$ , suggesting a rounding error or missing data.

- **[writing]** Figure 9: The text labels for the inner ring segments (e.g., ‘Point Goal’, ‘Person Following’) are rotated and difficult to read, reducing the clarity of the data breakdown.
- **[writing]** Figure 10: The caption describes the ‘Right’ panel as having ‘affordance pixel annotation’, but the image labels the green dot as ‘Affordance Pixel’ and the red dot as ‘Target’/‘Perturbed Target’ without explicitly defining the distinction in the caption text.
- **[writing]** Figure 10: The ‘Structured Sample’ panel includes a ‘Mission’ text block with coordinates, but the caption does not explain the coordinate system or the meaning of the values.
- **[science]** Figure 11: The caption describes a ‘three-stage pipeline’ and ‘generates and verifies affordance and target pixels,’ but the visual diagram only shows ‘VLM Instruction Decomposition’ and ‘VLM Annotation-Align’ without explicitly depicting the verification step or the generation of target pixels (which are listed as empty in the sample).
- **[writing]** Figure 11: The ‘Structured Sample’ on the right lists ‘Target Pixel: left:[], front:[], right[]’ as empty, which contradicts the caption’s claim that the sample shows ‘pixel-level annotations for affordance and target’.
- **[science]** Figure 12: The caption claims the pipeline uses ‘A\* consistency filtering’, but the diagram labels the filter simply as ‘Consistency Filter’ without explicitly showing or labeling the A\* algorithm component.
- **[writing]** Figure 12: The ‘Structured Sample’ text block contains raw JSON-like syntax (e.g., ‘left:[], front:[542, 698]’) which is difficult to read and visually cluttered compared to the rest of the figure.

## Jargon Police

*Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_jargon\_police.md

The paper introduces a sophisticated architecture and training pipeline, but it relies on several acronyms and technical terms that are not defined at their first occurrence, creating barriers for a competent reader from an adjacent field (e.g., a robotics researcher less familiar with specific VLM training nuances).

Specifically, the abstract and introduction use “CoT” (Chain-of-Thought) and “GRPO” (Group Relative Policy Optimization) without expansion. While these are becoming common in the LLM subfield, they are not universal across robotics or computer vision. Similarly, Section 3.1 introduces “SE(2)” without defining it as the Special Euclidean group in 2D, which is

standard in robotics but should be explicit for a general VLN audience. Section 4.2 mentions “Dagger” (Dataset Aggregation) and “smooth- $L_1$ ” loss without brief context, assuming the reader knows these specific algorithmic names and loss properties. Finally, Section 5.1 uses “KL” for Kullback-Leibler divergence without expansion.

These are minor omissions that can be fixed with simple parenthetical expansions at first use. They do not reflect a lack of rigor but rather an assumption of shared vocabulary that excludes the adjacent-field reader the paper aims to reach. Addressing these will make the paper self-contained and accessible without diluting its technical precision.

#### Action items.

- **[writing]** Section 1 (Introduction) uses ‘CoT’ and ‘GRPO’ without definition. Expand to ‘Chain-of-Thought (CoT)’ and ‘Group Relative Policy Optimization (GRPO)’ at first use.
- **[writing]** Section 3.1 (Preliminaries) uses ‘SE(2)’ without definition. Add ‘SE(2) (Special Euclidean group in 2D)’ at first occurrence.
- **[writing]** Section 4.2 (Pretraining Data Recipe) uses ‘Dagger’ without definition. Add ‘Dataset Aggregation (Dagger)’ at first mention.
- **[writing]** Section 4.2 uses ‘smooth- $L_1$ ’ without defining the loss function or its properties in this context. Add a brief clause explaining it is a robust regression loss.
- **[writing]** Section 5.1 (Post-Training) uses ‘KL’ without defining it as Kullback-Leibler divergence. Expand at first use.

## Logical Consistency

*Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_logical\_consistency.md

The paper’s logical structure is generally sound: the slow-fast architecture is consistently defined as a solution to coordinate drift and interpretability issues, and the experimental setup aligns with these definitions. The five tasks and the pixel-goal interface are used consistently from the Introduction through the Methods.

However, there is a critical numerical inconsistency in the Experiments section that breaks the internal argument:

In Section 6.1.5 (“Cross-Task Summary”), the text states: “ABot-N1 improves... by +0.558 absolute jump in  $SR_{<2m}$  on ABotN-POIBench.” This claim directly contradicts the data presented in Table 6 and the Abstract, which report the Success Rate (SR) increasing from 42.3% to 77.3%. The actual difference is 35.0 percentage points (0.350), not 0.558. The value 0.558 does not correspond to any obvious calculation derived from the reported numbers (e.g., it is not the ratio  $77.3/42.3$ , nor the difference  $0.773 - 0.423$ ).

This creates a non-sequitur where the summary conclusion does not follow from the evidence



table provided in the same paper. The argument that the model achieves “massive gains” is supported by the table, but the specific quantitative claim in the text is mathematically inconsistent with that table. This must be corrected to ensure the paper’s conclusions follow from its own presented data.

#### Action items.

- **[writing]** Section 6.1.5 claims a ‘+0.558 absolute jump’ in POI SR, but Table 6 and the Abstract show a rise from 42.3% to 77.3% (a 35.0 point jump). The value 0.558 contradicts the paper’s own data. Correct the text to match the table (e.g., ‘+35.0 points’).

## Overreach

*Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_overreach.md

The paper makes several broad claims about the generality, robustness, and completeness of the ABot-N1 model that exceed the scope of the provided evidence.

First, the **Abstract** and **Conclusion** assert that the model “ensures robust, generalizable, and interpretable navigation across simulation and real-world benchmarks.” While the simulation results are extensive, the “real-world” evidence is limited to qualitative case studies (Section 6.2, Figures 6-10) on a single quadrupedal robot platform. No quantitative metrics (Success Rate, SPL) are reported for the real-world deployment, nor is the method tested on different robot embodiments. The claim of “ensuring” robustness across real-world benchmarks is not licensed by qualitative examples alone. The text should be hedged to reflect that real-world generalization is *demonstrated qualitatively* on a specific platform, or quantitative real-world results must be added.

Second, the **Abstract** uses the strong verb “solves” regarding challenges like “coordinate drift and poor handling of long-tail semantics.” The experimental results (Section 6.1) show significant *improvements* (e.g., a 35% boost in POI arrival, reducing failure rates from ~58% to ~23% in some cases), but they do not eliminate these problems. A 4.6% failure rate on indoor point-goal navigation (Table 4) indicates the problem is not “solved.” The language should be adjusted to “significantly mitigates” or “substantially reduces” to accurately reflect the residual error rates.

Third, the **Introduction** and **Conclusion** frame ABot-N1 as a “general” foundation model for “open-world environments.” The evaluation is strictly confined to five specific navigation tasks (point-goal, object-goal, POI-goal, instruction-following, person-following) on a fixed set of benchmarks (R2R-CE, RxR-CE, OVON, EVT-Bench, and two new internal benchmarks). There is no evidence of zero-shot transfer to entirely new task types (e.g., exploration, manipulation) or performance in truly open-world, unstructured environments outside the training distribution. The claim of “generality” should be scoped to “generalist within the five evaluated navigation tasks” or supported by broader generalization experiments.



Finally, the **Conclusion** states the model “achieves state-of-the-art results on all five benchmarks.” While the model leads on most metrics, Table 1 shows that on the RxR-CE Val-Unseen split, ABot-N1’s Success Rate (73.9%) is second to Qwen-RobotNav-4B (75.2%). The summary statement is therefore factually imprecise regarding the “all five” claim if interpreted as leading on every metric of every benchmark. The text should clarify that it achieves SOTA on the *primary* metrics of the tasks or specify the exact metrics where it leads.

These issues are primarily matters of rhetorical scope and honesty about limitations. They can be resolved by tightening the language in the Abstract and Conclusion to match the specific boundaries of the experimental evidence.

#### Action items.

- **[writing]** Abstract claims the method ‘ensures robust... navigation across... real-world benchmarks,’ but Section 6.2 only provides qualitative case studies on one robot. No quantitative real-world metrics exist. Replace ‘ensures... across real-world benchmarks’ with ‘demonstrates qualitative robustness on a single quadrupedal platform’ or add quantitative real-world evaluation.
- **[writing]** Abstract states the method ‘solves’ coordinate drift and long-tail semantics. Results show significant reduction (e.g., 35% POI boost) but not elimination (4.6% indoor failure remains). Change ‘solves’ to ‘significantly mitigates’ to align with residual error rates.
- **[writing]** Introduction/Conclusion claim a ‘general’ model for ‘open-world environments,’ yet experiments cover only five specific tasks on fixed benchmarks. No zero-shot transfer to new tasks or unstructured domains is shown. Narrow the claim to ‘generalist within the five evaluated tasks’ or provide broader generalization evidence.
- **[writing]** Conclusion states the model ‘achieves state-of-the-art results on all five benchmarks.’ However, Table 1 shows RxR-CE SR (73.9%) is second to Qwen-RobotNav-4B (75.2%). Clarify that SOTA is achieved on primary metrics or specify which metrics lead, as the current claim is factually inaccurate for RxR SR.

## Safety Ethics

*Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_safety\_ethics.md

This paper presents a visual-language navigation foundation model (ABot-N1) and associated benchmarks for embodied agents. From a safety and ethics perspective, the work is low-risk. The research focuses on improving navigation robustness, interpretability (via Chain-of-Thought), and social compliance (avoiding vehicle lanes, respecting pedestrian zones) in simulated and real-world environments.

The paper explicitly addresses safety as a core design goal, introducing a “Safety Clearance” reward function in its GRPO post-training stage to penalize proximity to non-traversable

regions and traffic-rule violations (Section 4.3.2). The benchmarks (ABotN-PointBench, ABotN-POIBench) are constructed using high-fidelity 3DGS reconstructions of real-world scenes, but the data collection and annotation processes described (LiDAR-inertial SLAM, manual annotation of walkable regions) do not appear to involve the collection of personally identifiable information (PII) or sensitive human data in a way that would require IRB approval for the *navigation* task itself. The “person-following” task uses synthetic avatars in simulation (Habitat) or teleoperation data where the focus is on tracking behavior, not biometric identification, and the paper does not claim to release raw video of identifiable individuals.

There are no indications of dual-use capabilities intended for surveillance, deception, or autonomous weaponization. The “social compliance” features are designed to prevent harm to humans and property, not to facilitate covert operations. The release of benchmarks and code is framed as advancing the field of safe navigation. No specific, non-trivial risks of harm are identified that are unacknowledged or unmitigated in the text. The paper does not require additional safety disclosures or mitigations beyond what is already present.

## Scientific Evidence

*Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_scientific\_evidence.md

The paper presents a compelling architecture and reports impressive numerical gains across five navigation tasks. However, the experimental design currently lacks the rigor required to definitively attribute these gains to the proposed mechanisms (slow-fast factorization, pixel goals, GRPO) rather than confounding factors like training scale, data volume, or random seed variance.

First, the evidence for “state-of-the-art” performance relies entirely on single-run results reported in Tables 1 through 5. There is no reporting of standard deviation, confidence intervals, or the number of random seeds used for training and evaluation. In embodied navigation, performance can vary significantly based on initialization and stochastic rollout dynamics. A 1-2% gain (e.g., R2R-CE SR 70.9% vs 69.3%) or even a larger gain (e.g., +35% on POI) could plausibly arise from a lucky seed or a favorable test split without the proposed method being fundamentally superior. To support the claim of robust improvement, the authors must report results averaged over at least 3-5 independent seeds with variance metrics.

Second, the attribution of performance gains to specific architectural components is weak due to missing ablations. The paper compares ABot-N1 primarily against ABot-N0. Since ABot-N0 is the authors’ own prior work, it likely shares the same data engine, pre-training corpus, and potentially similar architectural backbones. The reported improvements (e.g., +16% SR on PointBench) could be driven by the increased scale of the data (30M samples), the specific GRPO post-training, or the pixel-goal interface, but the current design does not isolate these variables. For instance, there is no experiment showing a version of ABot-N1 trained with the

same data and GRPO but using a latent-vector interface instead of pixel goals. Without such controls, the claim that the “pixel goal” or “slow-fast” design is the primary driver of success remains an unproven hypothesis.

Finally, the introduction of new benchmarks (ABotN-PointBench, ABotN-POIBench) introduces a potential confound. The paper claims SOTA performance on these benchmarks, but it does not demonstrate that existing strong baselines (like ViNT or NoMaD) fail on these tasks due to the specific challenges (social compliance, POI entrance precision) rather than simply because they were not fine-tuned on the new data. If baselines were given the same amount of tuning effort on the new datasets, the performance gap might narrow significantly. The authors should report a fair comparison where strong baselines are re-trained or fine-tuned on the new benchmarks to ensure the “SOTA” claim reflects genuine architectural superiority rather than benchmark novelty.

### Action items.

- **[writing]** The paper presents a compelling architecture and reports impressive numerical gains across five navigation tasks. However, the experimental design currently lacks the rigor required to definitively attribute these gains to the proposed mechanisms (slow-fast factorization, pixel goals, GRPO) rather than confounding factors like training scale, data volume, or random seed variance. First, the evidence for “state-of-the-art” performance relies entirely on single-run results reported in Tables 1 thr

## Statistical Analysis

*Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_statistical\_analysis.md

The statistical reporting in the experimental section is currently insufficient to support the paper’s strong claims of “state-of-the-art” performance and “massive gains.” The primary issue is the complete absence of uncertainty quantification. Tables 1 through 5 present single-point estimates (e.g., “92.9% SR” in Table 4, “77.3% SR” in Table 5) derived from what appears to be single runs or unreported aggregations. In the context of deep learning and embodied AI, where performance can vary significantly due to random seeds, initialization, or stochastic environment interactions, a single number provides no information about the stability or reliability of the result. The field standard for such claims is to report the mean and standard deviation (or standard error) over multiple independent runs (typically 3-5 seeds). Without this, a reader cannot judge if the reported 35% gain in POI-goal navigation is a robust improvement or a lucky run.

Furthermore, the paper makes frequent use of the term “significant” (e.g., “statistically significant” is implied by the magnitude of gains and the “SOTA” label) without providing the necessary statistical machinery to back it up. There are no p-values, confidence intervals, or effect sizes reported. When comparing against baselines, the authors should either perform a

formal statistical test (e.g., a paired t-test if the same random seeds were used for both models, or a bootstrap test) or at minimum provide the variance across seeds to allow for a visual or heuristic assessment of overlap.

Finally, the analysis stratifies results by difficulty tiers (Low/Medium/High) and environment types (Indoor/Outdoor) across multiple tasks. This creates a large number of pairwise comparisons. Highlighting specific sub-splits where the model performs best without correcting for multiple comparisons (e.g., using Bonferroni or False Discovery Rate control) inflates the risk of Type I errors (false positives). The current presentation risks cherry-picking the most favorable sub-splits to claim superiority. To rectify this, the authors should either apply a multiple-comparison correction to their significance claims or rephrase the results to describe the observed trends without invoking statistical significance.

#### Action items.

- **[writing]** Tables 1-5 report single-point performance metrics (e.g., ‘92.9% SR’) without any measure of uncertainty (standard deviation, standard error, or confidence intervals) across multiple random seeds or runs. In deep learning, single-run results are insufficient to distinguish signal from stochastic variance. Report mean  $\pm$  SD over at least 3-5 seeds for all primary results, or explicitly state that results are from a single run and treat them as such.
- **[writing]** The abstract and Section 6 claim ‘state-of-the-art’ and ‘massive gains’ (e.g., ‘+35.0% boost’) based on point estimates. Without reported variance or statistical significance tests (e.g., paired t-tests or bootstrap CIs) comparing ABot-N1 against the strongest baselines, it is impossible to determine if these differences are statistically significant or within the noise floor of the evaluation protocol. Add uncertainty bounds or significance markers to the tables.
- **[writing]** Section 6.1.3 and 6.1.4 report results stratified by difficulty tiers (Low, Medium, High) and split by indoor/outdoor. This constitutes multiple hypothesis testing (e.g., 3 tiers  $\times$  2 splits  $\times$  2 metrics = 12+ comparisons per task). The paper highlights specific ‘significant’ improvements in these sub-splits without applying a correction for multiple comparisons (e.g., Bonferroni or Holm). Re-evaluate significance claims with a correction or explicitly acknowledge the multiplicity risk.

## Writing Quality

*Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_writing\_quality.md

The manuscript presents a complex architecture and extensive experimental results, but the prose occasionally struggles to deliver these details with the necessary clarity and momentum. While the technical content is dense, the writing quality generally allows a reader to follow the argument, though there are specific instances where sentence structure and paragraph

organization impede smooth reading.

The most significant friction points occur in the Methods section, particularly where the authors introduce complex mathematical justifications for their training strategies. In Section 4.3, the transition into the derivation of the sampling ratio for GRPO is abrupt. The text jumps immediately into a formal analysis of variance and Taylor expansions without a clear topic sentence to orient the reader to the *purpose* of this derivation. A reader must work to infer that this math is intended to justify the 5:3:2 sampling ratio, rather than being an abstract theoretical exercise. A simple lead-in sentence summarizing the goal of the analysis would significantly improve the flow.

Similarly, in Section 4.2, the explanation of why the smooth-L1 loss does not cause mode collapse in this specific architecture is buried within a long, multi-clause sentence. The sentence structure forces the reader to hold the “problem” (mode collapse in generic VLN) and the “solution” (dual-system factorization) in working memory simultaneously before the connection is made. Breaking this into two sentences—one stating the general problem and the other explaining the specific architectural fix—would make the logic more immediate and easier to parse on the first pass.

In the Results section, the “Analysis” paragraphs occasionally suffer from slightly awkward phrasing that slows down the reader. For instance, the comparison between the multi-task model and the single-task variant in Section 5.1 uses a colon in a way that feels grammatically disjointed. While the meaning is recoverable, a more standard phrasing would eliminate the need for the reader to pause and re-parse the sentence structure.

Finally, the Conclusion attempts to summarize the entire paper’s contribution in a single, massive sentence. This “kitchen sink” approach dilutes the impact of the key findings. Splitting this into distinct sentences for the architecture, the results, and the real-world validation would make the conclusion more punchy and readable.

Overall, the paper is well-structured, but tightening the sentence construction in the Methods and Results sections, and ensuring that complex derivations are properly introduced, would elevate the writing from “understandable” to “effortless.”

#### **Action items.**

- **[writing]** The manuscript presents a complex architecture and extensive experimental results, but the prose occasionally struggles to deliver these details with the necessary clarity and momentum. While the technical content is dense, the writing quality generally allows a reader to follow the argument, though there are specific instances where sentence structure and paragraph organization impede smooth reading. The most significant friction points occur in the Methods section, particularly where the autho